# "Inner product"

$v = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_d \end{pmatrix}$ $u = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_d \end{pmatrix}$

Inner product (generalization of the dot product) takes two vectors $v, u \in V$ and provides a scalar $\langle v|u\rangle$ that belongs to the field $F$.

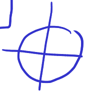
$F = \mathbb{R}$

$|\alpha_i|v\rangle = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_d \end{pmatrix}$

bra: $\langle v| = (\alpha_1^* \cdots \alpha_d^*)$

$= v^T u =$

$$\underbrace{\langle v|u\rangle}_{\text{braket notation}} = v^\dagger u = \begin{pmatrix} \alpha_1^* & \cdots & \alpha_d^* \end{pmatrix} \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_d \end{pmatrix} = \boxed{\sum_{i=1}^{d} \alpha_i^* \beta_i}$$



**Conjugate symmetry:** $\langle v|u\rangle = \langle u|v\rangle^*$ $\Rightarrow$ $v^T u = u^T v$.

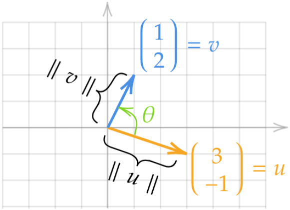
**Length (norm) of a vector:** $\| v \|_2 = \sqrt{\langle v|v\rangle} = \sqrt{\sum_{i=1}^{d} |\alpha_i|^2}$ (normalized vector $\| v \| = 1$)

$\ell_p$-norm $= \left( \sum_{i=0}^{d} |\alpha_i|^p \right)^{1/p}$

**Angle between two real vectors:** $\boxed{\cos\theta = \frac{\langle v|u\rangle}{\|v\|\,\|u\|}}$ (orthogonal vectors $\langle v|u\rangle = 0$)

$\| v \| = \sqrt{5}$ and $\| u \| = \sqrt{10}$ , $\ell_2$ norm

**Unit vector:** $\hat{v} = \frac{v}{\|v\|} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$




video

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